

One-Step Method

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First Order ODEs: $y' = f(x, y)$

Initial Condition: $y(x_i) = y_i$

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Explicit Runge-Kutta Methods

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Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^s a_{j,m} k_m\right)$$

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}

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$

→

c_1	$a_{1,1}$	$a_{1,2}$	$\cdots$	$a_{1,s-1}$	$a_{1,s}$
c_2	$a_{2,1}$	$a_{2,2}$	$\cdots$	$a_{2,s-1}$	$a_{2,s}$
c_3	$a_{3,1}$	$a_{3,2}$	$\cdots$	$a_{3,s-1}$	$a_{3,s}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
c_s	$a_{s,1}$	$a_{s,2}$	$\cdots$	$a_{s,s-1}$	$a_{s,s}$
	b_1	b_2	$\cdots$	b_{s-1}	b_s

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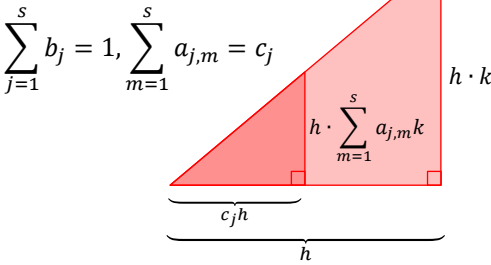
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Explicit Runge-Kutta Methods

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c_1	$a_{1,1}$	$a_{1,2}$	$\cdots$	$a_{1,s-1}$	$a_{1,s}$
c_2	$a_{2,1}$	$a_{2,2}$	$\cdots$	$a_{2,s-1}$	$a_{2,s}$
c_3	$a_{3,1}$	$a_{3,2}$	$\cdots$	$a_{3,s-1}$	$a_{3,s}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
c_s	$a_{s,1}$	$a_{s,2}$	$\cdots$	$a_{s,s-1}$	$a_{s,s}$
	b_1	b_2	$\cdots$	b_{s-1}	b_s

$y(x) = y_i + k \cdot (x - x_i) \rightarrow k_1 = k_2 = k_m = \cdots = k$
 $y_{i+1} = y_i + k \cdot (x_{i+1} - x_i)$
 $y_{i+1} = y_i + h \cdot k$



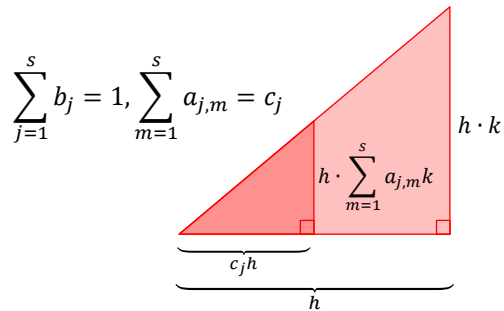
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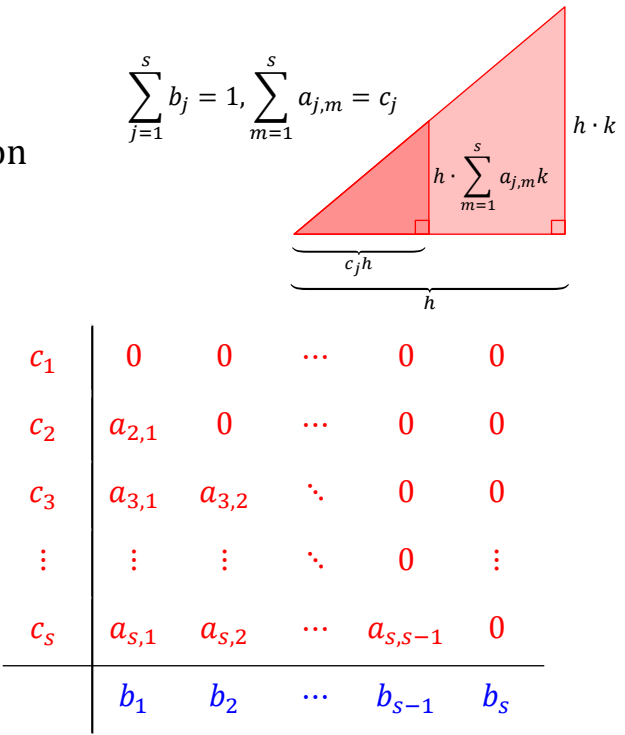
Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^s a_{j,m} k_m\right)$$
$$y_{i+1} = y_i + h \cdot \sum_{j=1}^s b_j k_j$$

$$\left. \vphantom{\sum_{j=1}^s} \right\}$$

$\mathbf{c}$	$\mathbf{A}$
	$\mathbf{b}^T$

 $\rightarrow$



One-Step Method

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Second Order Explicit Runge-Kutta Methods

One-Step Method

Second Order Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^2 a_{j,m} k_m\right)$$

One-Step Method

Second Order Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + \textcolor{red}{c}_j h, y_i + h \cdot \sum_{m=1}^2 \textcolor{red}{a}_{j,m} k_m\right)$$

$$y_{i+1} = y_i + h \cdot \sum_{j=1}^2 \textcolor{blue}{b}_j k_j$$

One-Step Method

Second Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + \mathbf{c}_j h, y_i + h \cdot \sum_{m=1}^2 \mathbf{a}_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^2 \mathbf{b}_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$

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Second Order Explicit Runge-Kutta Methods

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$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2)$$

One-Step Method

Second Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^2 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^2 b_j k_j \end{aligned} \right\} \quad \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cc} & & \\ \hline c_1 & 0 & 0 \\ c_2 & a_{2,1} & 0 \\ \hline & b_1 & b_2 \end{array}$$

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2)$$

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Second Order Explicit Runge-Kutta Methods

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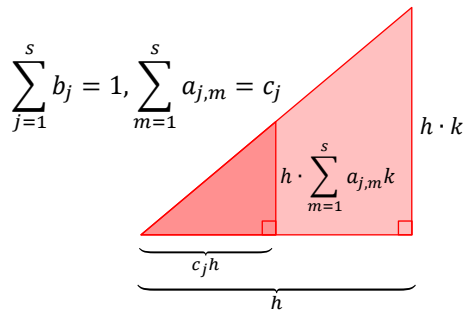
$$y_{i+1} = y_i + h \cdot (b_1 k_1 + b_2 k_2)$$

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Second Order Explicit Runge-Kutta Methods

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$$\begin{aligned} k_1 &= f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2) \\ k_2 &= f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1 + h \cdot 0 k_2) \\ y_{i+1} &= y_i + h \cdot (b_1 k_1 + b_2 k_2) \end{aligned}$$



One-Step Method

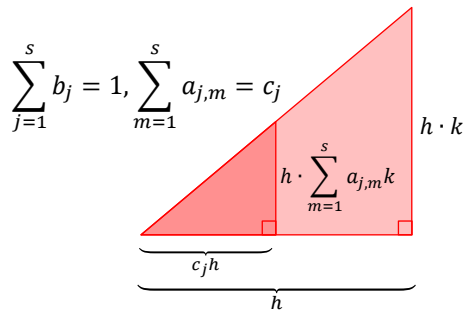
Second Order Explicit Runge-Kutta Methods

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$$k_1 = f(x_i, y_i)$$

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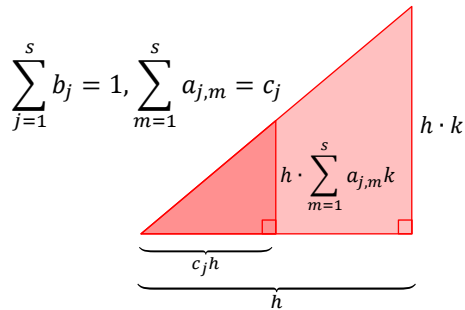
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$$y_{i+1} = y_i + h \cdot (b_1 k_1 + b_2 k_2)$$



$$b_1 + b_2 = 1$$

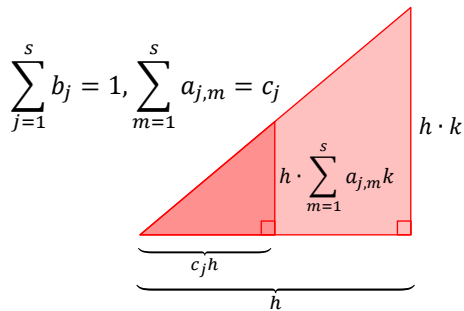
$$b_2 \cdot c_2 = \frac{1}{2}$$

$$b_2 \cdot a_{2,1} = \frac{1}{2}$$

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Second Order Explicit Runge-Kutta Methods

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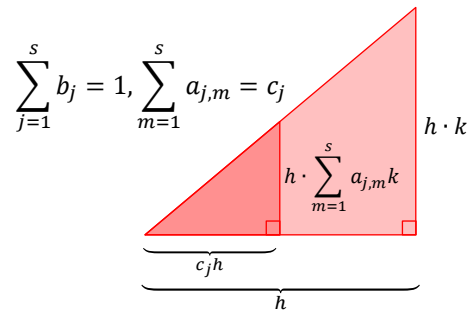
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One-Step Method

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The Midpoint Method:



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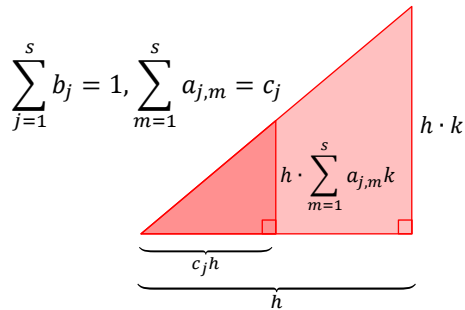
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Second Order Explicit Runge-Kutta Methods

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The Midpoint Method: $b_1 = 0, b_2 = 1, c_2 = a_{2,1} = \frac{1}{2}$



$$b_1 + b_2 = 1$$

$$b_2 \cdot c_2 = \frac{1}{2}$$

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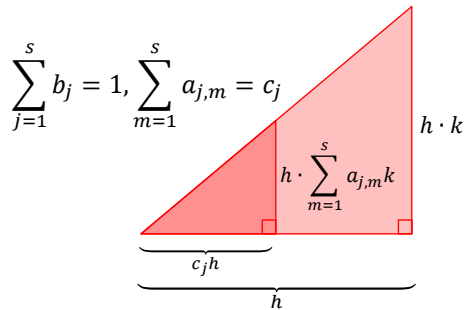
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$$k_1 = f(x_i, y_i)$$



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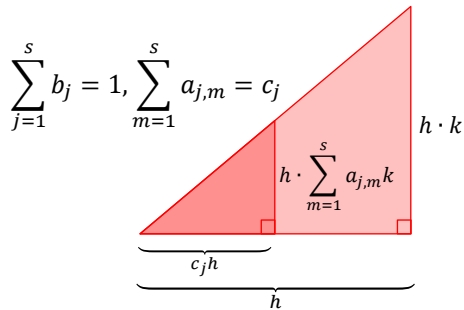
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$$k_2 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right)$$



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One-Step Method

Second Order Explicit Runge-Kutta Methods

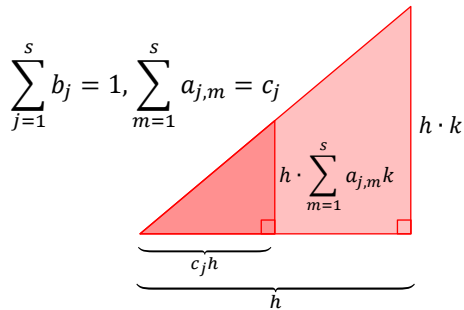
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$$b_1 + b_2 = 1$$

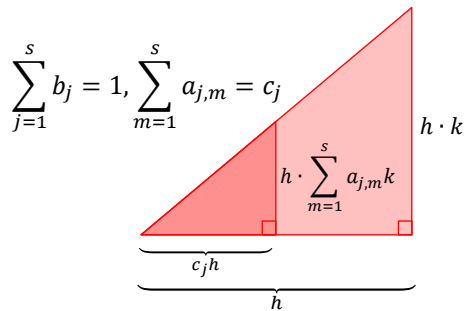
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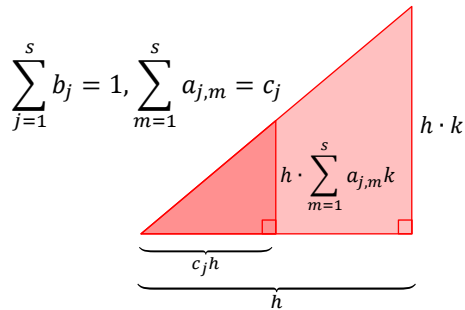
$$b_2 \cdot a_{2,1} = \frac{1}{2}$$

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Modified Euler Method:



$$b_1 + b_2 = 1$$

$$b_2 \cdot c_2 = \frac{1}{2}$$

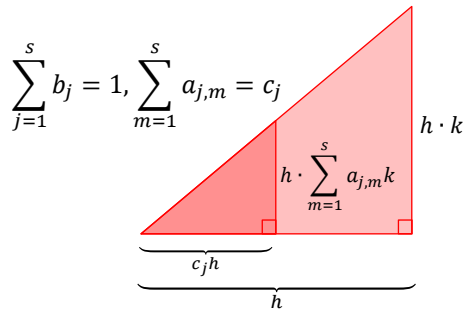
$$b_2 \cdot a_{2,1} = \frac{1}{2}$$

One-Step Method

Second Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^2 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^2 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cc} c_1 & 0 & 0 \\ c_2 & a_{2,1} & 0 \\ \hline & b_1 & b_2 \end{array}$$

Modified Euler Method: $b_1 = \frac{1}{2}, b_2 = \frac{1}{2}, c_2 = a_{2,1} = 1$



$$b_1 + b_2 = 1$$

$$b_2 \cdot c_2 = \frac{1}{2}$$

$$b_2 \cdot a_{2,1} = \frac{1}{2}$$

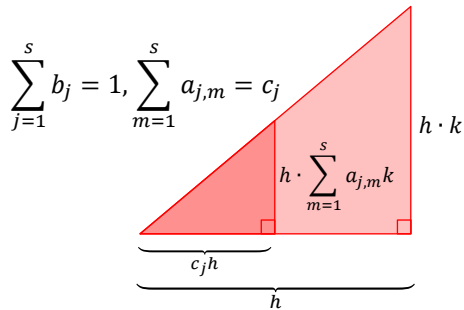
One-Step Method

Second Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^2 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^2 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cc} c_1 & 0 & 0 \\ c_2 & a_{2,1} & 0 \\ \hline & b_1 & b_2 \end{array}$$

Modified Euler Method: $b_1 = \frac{1}{2}, b_2 = \frac{1}{2}, c_2 = a_{2,1} = 1$

$$k_1 = f(x_i, y_i)$$



$$b_1 + b_2 = 1$$

$$b_2 \cdot c_2 = \frac{1}{2}$$

$$b_2 \cdot a_{2,1} = \frac{1}{2}$$

One-Step Method

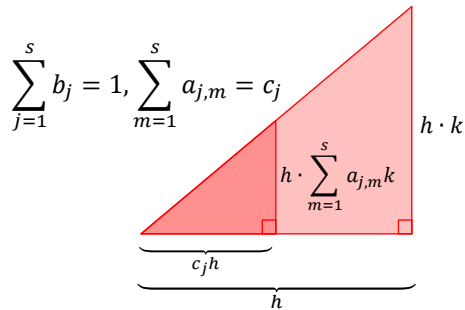
Second Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^2 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^2 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cc} c_1 & 0 & 0 \\ c_2 & a_{2,1} & 0 \\ \hline & b_1 & b_2 \end{array}$$

Modified Euler Method: $b_1 = \frac{1}{2}, b_2 = \frac{1}{2}, c_2 = a_{2,1} = 1$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + h, y_i + h k_1)$$



$$b_1 + b_2 = 1$$

$$b_2 \cdot c_2 = \frac{1}{2}$$

$$b_2 \cdot a_{2,1} = \frac{1}{2}$$

One-Step Method

Second Order Explicit Runge-Kutta Methods

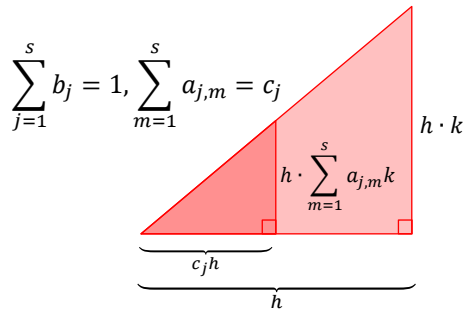
$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^2 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^2 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cc} c_1 & 0 & 0 \\ c_2 & a_{2,1} & 0 \\ \hline & b_1 & b_2 \end{array}$$

Modified Euler Method: $b_1 = \frac{1}{2}, b_2 = \frac{1}{2}, c_2 = a_{2,1} = 1$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + h, y_i + h k_1)$$

$$y_{i+1} = y_i + \frac{h}{2} \cdot (k_1 + k_2)$$



$$b_1 + b_2 = 1$$

$$b_2 \cdot c_2 = \frac{1}{2}$$

$$b_2 \cdot a_{2,1} = \frac{1}{2}$$

One-Step Method

One-Step Method

Third Order Explicit Runge-Kutta Methods

One-Step Method

Third Order Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right)$$

One-Step Method

Third Order Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + \textcolor{red}{c}_j h, y_i + h \cdot \sum_{m=1}^3 \textcolor{red}{a}_{j,m} k_m\right)$$

$$y_{i+1} = y_i + h \cdot \sum_{j=1}^3 \textcolor{blue}{b}_j k_j$$

One-Step Method

Third Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + \mathbf{c}_j h, y_i + h \cdot \sum_{m=1}^3 \mathbf{a}_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^3 \mathbf{b}_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$

One-Step Method

Third Order Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right)$$
$$y_{i+1} = y_i + h \cdot \sum_{j=1}^3 b_j k_j$$

}

$\mathbf{c}$	$\mathbf{A}$
	$\mathbf{b}^T$

$\rightarrow$

c_1	0	0	0
c_2	$a_{2,1}$	0	0
c_3	$a_{3,1}$	$a_{3,2}$	0
	b_1	b_2	b_3

One-Step Method

Third Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^3 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|ccc} & & & \\ \hline c_1 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}$$

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

One-Step Method

Third Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^3 b_j k_j \end{aligned} \right\} \quad \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \quad \rightarrow \quad \begin{array}{c|ccc} & & & \\ \hline c_1 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}$$

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

One-Step Method

Third Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^3 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|ccc} & & & & \\ \hline c_1 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}$$

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1} k_1 + h \cdot a_{3,2} k_2 + h \cdot 0 k_3)$$

One-Step Method

Third Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^3 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|ccc} & c_1 & c_2 & c_3 \\ \hline c_1 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}$$

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1} k_1 + h \cdot a_{3,2} k_2 + h \cdot 0 k_3)$$

$$y_{i+1} = y_i + h \cdot (b_1 k_1 + b_2 k_2 + b_3 k_3)$$

One-Step Method

Third Order Explicit Runge-Kutta Methods

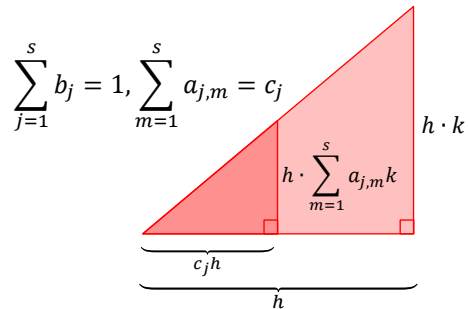
$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^3 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cc} c_1 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}$$

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1 + h \cdot 0 k_2 + h \cdot 0 k_3)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1} k_1 + h \cdot a_{3,2} k_2 + h \cdot 0 k_3)$$

$$y_{i+1} = y_i + h \cdot (b_1 k_1 + b_2 k_2 + b_3 k_3)$$



One-Step Method

Third Order Explicit Runge-Kutta Methods

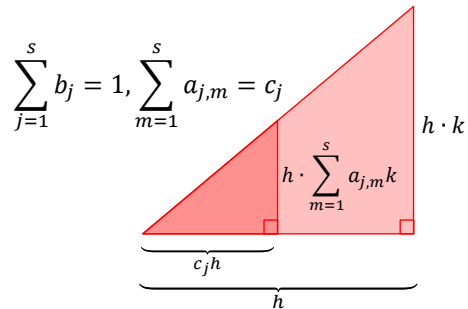
$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^3 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cc} c_1 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}$$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1} k_1 + h \cdot a_{3,2} k_2)$$

$$y_{i+1} = y_i + h \cdot (b_1 k_1 + b_2 k_2 + b_3 k_3)$$



One-Step Method

Third Order Explicit Runge-Kutta Methods

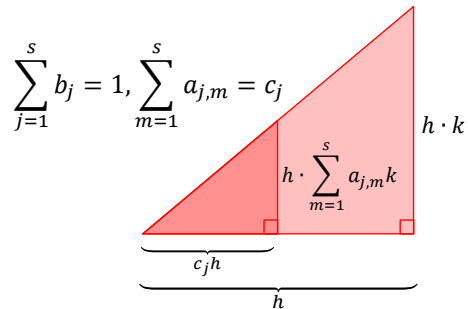
$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^3 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^3 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cc} c_1 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}$$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(x_i + h, y_i - hk_1 + 2hk_2\right)$$

$$y_{i+1} = y_i + \frac{h}{6} \cdot (k_1 + 4k_2 + k_3)$$



One-Step Method

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right)$$

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$k_j = f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right)$$

$$y_{i+1} = y_i + h \cdot \sum_{j=1}^4 b_j k_j$$

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + \mathbf{c}_j h, y_i + h \cdot \sum_{m=1}^4 \mathbf{a}_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 \mathbf{b}_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\}$$

c	A
	b^T

 $\rightarrow$

c_1	0	0	0	0
c_2	$a_{2,1}$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	0
	b_1	b_2	b_3	b_4

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\}$$

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$



c_1	0	0	0	0
c_2	$a_{2,1}$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	0
	b_1	b_2	b_3	b_4

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0 k_1 + h \cdot 0 k_2 + h \cdot 0 k_3 + h \cdot 0 k_4)$$

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cccc} c_1 & 0 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 & 0 \\ c_4 & a_{4,1} & a_{4,2} & a_{4,3} & 0 \\ \hline & b_1 & b_2 & b_3 & b_4 \end{array}$$

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\}$$

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$



c_1	0	0	0	0
c_2	$a_{2,1}$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	0
	b_1	b_2	b_3	b_4

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1} k_1 + h \cdot a_{3,2} k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\}$$

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$



c_1	0	0	0	0
c_2	$a_{2,1}$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	0
	b_1	b_2	b_3	b_4

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1}k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1}k_1 + h \cdot a_{3,2}k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_4 = f(x_i + c_4 h, y_i + h \cdot a_{4,1}k_1 + h \cdot a_{4,2}k_2 + h \cdot a_{4,3}k_3 + h \cdot 0k_4)$$

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\} \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array} \rightarrow \begin{array}{c|cccc} c_1 & 0 & 0 & 0 & 0 \\ c_2 & a_{2,1} & 0 & 0 & 0 \\ c_3 & a_{3,1} & a_{3,2} & 0 & 0 \\ c_4 & a_{4,1} & a_{4,2} & a_{4,3} & 0 \\ \hline & b_1 & b_2 & b_3 & b_4 \end{array}$$

$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1} k_1 + h \cdot a_{3,2} k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_4 = f(x_i + c_4 h, y_i + h \cdot a_{4,1} k_1 + h \cdot a_{4,2} k_2 + h \cdot a_{4,3} k_3 + h \cdot 0k_4)$$

$$y_{i+1} = y_i + h \cdot (b_1 k_1 + b_2 k_2 + b_3 k_3 + b_4 k_4)$$

One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\}$$

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$



c_1	0	0	0	0
c_2	$a_{2,1}$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	0
	b_1	b_2	b_3	b_4

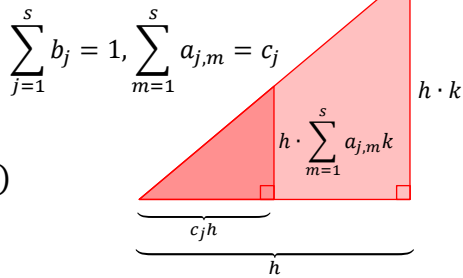
$$k_1 = f(x_i + c_1 h, y_i + h \cdot 0k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1}k_1 + h \cdot 0k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1}k_1 + h \cdot a_{3,2}k_2 + h \cdot 0k_3 + h \cdot 0k_4)$$

$$k_4 = f(x_i + c_4 h, y_i + h \cdot a_{4,1}k_1 + h \cdot a_{4,2}k_2 + h \cdot a_{4,3}k_3 + h \cdot 0k_4)$$

$$y_{i+1} = y_i + h \cdot (b_1 k_1 + b_2 k_2 + b_3 k_3 + b_4 k_4)$$



One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\}$$

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$



c_1	0	0	0	0
c_2	$a_{2,1}$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	0
	b_1	b_2	b_3	b_4

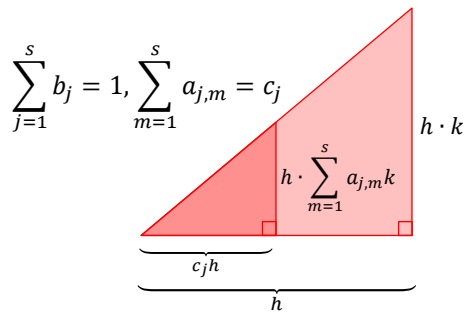
$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + c_2 h, y_i + h \cdot a_{2,1} k_1)$$

$$k_3 = f(x_i + c_3 h, y_i + h \cdot a_{3,1} k_1 + h \cdot a_{3,2} k_2)$$

$$k_4 = f(x_i + c_4 h, y_i + h \cdot a_{4,1} k_1 + h \cdot a_{4,2} k_2 + h \cdot a_{4,3} k_3)$$

$$y_{i+1} = y_i + h \cdot (b_1 k_1 + b_2 k_2 + b_3 k_3 + b_4 k_4)$$



One-Step Method

Fourth Order Explicit Runge-Kutta Methods

$$\left. \begin{aligned} k_j &= f\left(x_i + c_j h, y_i + h \cdot \sum_{m=1}^4 a_{j,m} k_m\right) \\ y_{i+1} &= y_i + h \cdot \sum_{j=1}^4 b_j k_j \end{aligned} \right\}$$

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$



c_1	0	0	0	0
c_2	$a_{2,1}$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	0
	b_1	b_2	b_3	b_4

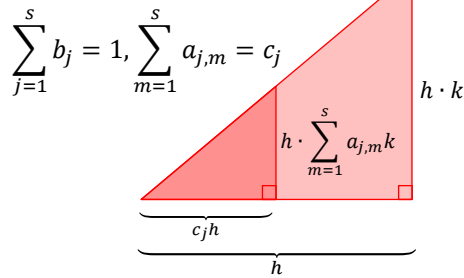
$$k_1 = f(x_i, y_i)$$

$$k_2 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_2\right)$$

$$k_4 = f(x_i + h, y_i + hk_3)$$

$$y_{i+1} = y_i + \frac{h}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4)$$



One-Step Method

One-Step Method

Domain of Stability

One-Step Method

Domain of Stability

Initial Value Problem: $y' = \lambda y; \quad y(0) = y_0 + \varepsilon_0$

One-Step Method

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Initial Value Problem: $y' = \lambda y; \quad y(0) = y_0 + \varepsilon_0 \quad \rightarrow \quad y(x_i) \approx y_i + \varepsilon_i$

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Second Order Methods:

$$\left| 1 + h\lambda + \frac{(h\lambda)^2}{2} \right| \leq 1$$

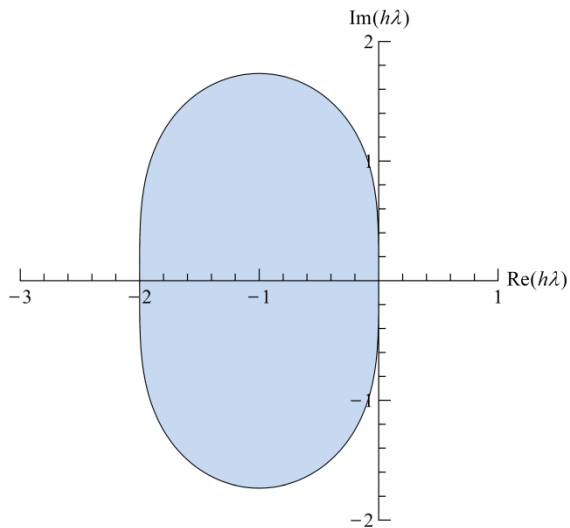
One-Step Method

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One-Step Method

Domain of Stability

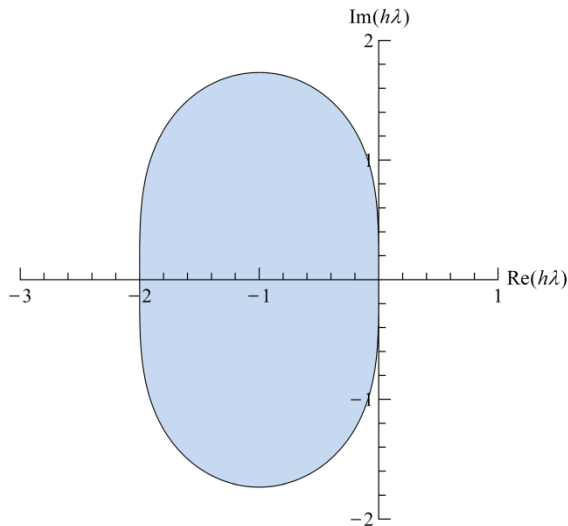
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Third Order Methods:

$$\left| 1 + h\lambda + \frac{(h\lambda)^2}{2} + \frac{(h\lambda)^3}{6} \right| \leq 1$$



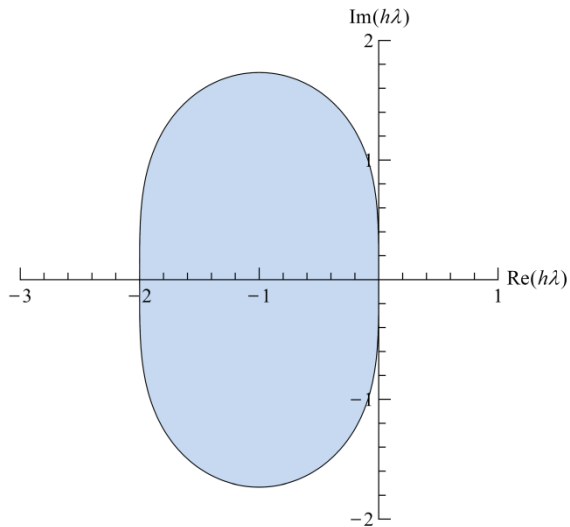
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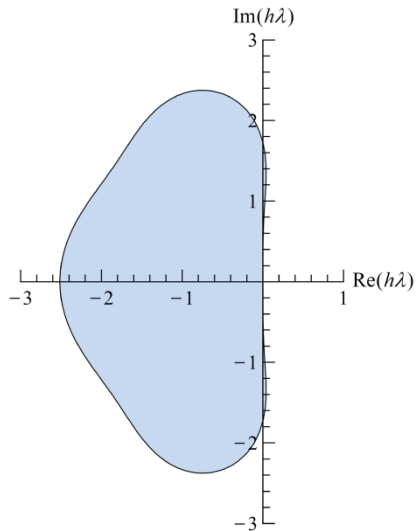
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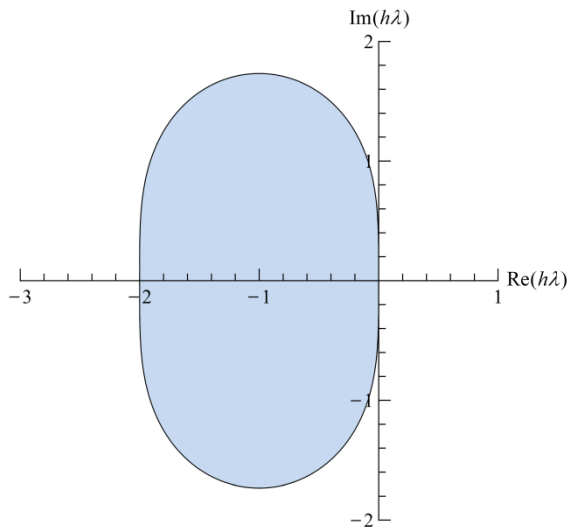
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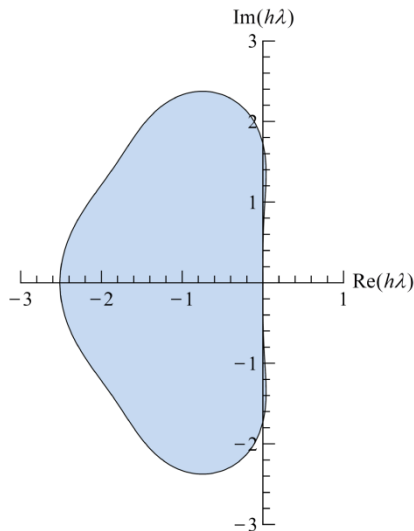
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Fourth Order Methods:

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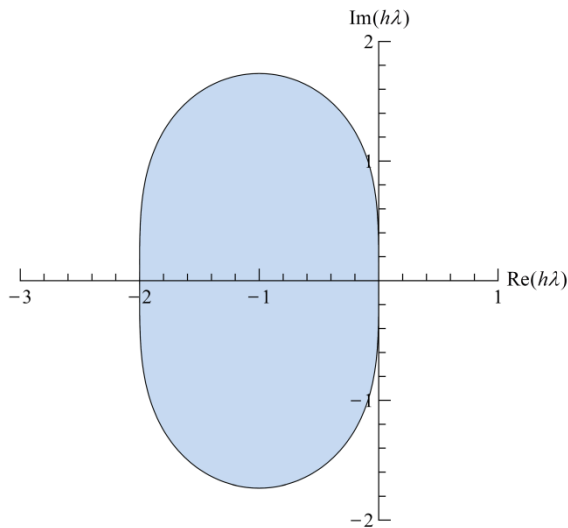
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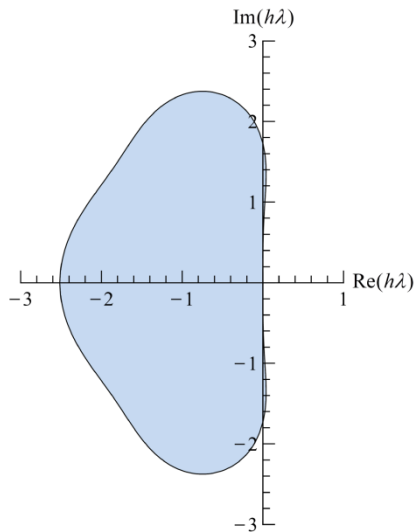
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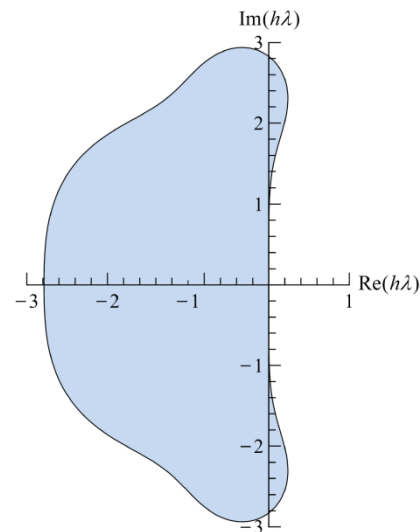
Third Order Methods:

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Fourth Order Methods:

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One-Step Method

Example 07-05: Approximate the numerical solution of the following initial value problem using **the midpoint method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2$$

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$$k_1 = f(x_i, y_i)$$

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$$\begin{aligned} i = 0: \quad k_1 &= f(x_0, y_0) = y_0 - x_0^2 + 1 \\ &= 0.5 - 0^2 + 1 = 1.5 \end{aligned}$$

One-Step Method

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$$y_{i+1} = y_i + h \cdot k_2$$

$i = 0:$ $k_1 = f(x_0, y_0) = y_0 - x_0^2 + 1$
 $= 0.5 - 0^2 + 1 = 1.5$

$$k_2 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_1\right) = \left(y_0 + \frac{h}{2}k_1\right) - \left(x_0 + \frac{h}{2}\right)^2 + 1$$
$$= \left(0.5 + \frac{0.2}{2} \cdot 1.5\right) - \left(0 + \frac{0.2}{2}\right)^2 + 1 = 1.64$$

One-Step Method

Example 07-05: Approximate the numerical solution of the following initial value problem using **the midpoint method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

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$$y_1 = y_0 + h \cdot k_2$$
$$= 0.5 + 0.2 \cdot 1.64 = 0.828$$

One-Step Method

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$i = 0:$ $k_1 = f(x_0, y_0) = y_0 - x_0^2 + 1$
 $= 0.5 - 0^2 + 1 = 1.5$

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$$y_1 = y_0 + h \cdot k_2$$
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$i = 1:$

One-Step Method

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 $= 0.5 - 0^2 + 1 = 1.5$

$$k_2 = f(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_1) = (y_0 + \frac{h}{2}k_1) - (x_0 + \frac{h}{2})^2 + 1$$
$$= (0.5 + \frac{0.2}{2} \cdot 1.5) - (0 + \frac{0.2}{2})^2 + 1 = 1.64$$

$$y_1 = y_0 + h \cdot k_2$$
$$= 0.5 + 0.2 \cdot 1.64 = 0.828$$

$i = 1:$ $k_1 = f(x_1, y_1) = y_1 - x_1^2 + 1$
 $= 0.828 - 0.2^2 + 1 = 1.788$

One-Step Method

Example 07-05: Approximate the numerical solution of the following initial value problem using **the midpoint method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2$$

$$\text{Exact Solution: } y(x) = (x + 1)^2 - \frac{1}{2}e^x$$

$$f(x, y) = y - x^2 + 1, N = \frac{2 - 0}{0.2} = 10, x_0 = 0, y_0 = 0.5$$

The Midpoint Method:

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1)$$

$$y_{i+1} = y_i + h \cdot k_2$$

$$\begin{aligned} i = 0: \quad k_1 &= f(x_0, y_0) = y_0 - x_0^2 + 1 \\ &= 0.5 - 0^2 + 1 = 1.5 \end{aligned}$$

$$\begin{aligned} k_2 &= f(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_1) = (y_0 + \frac{h}{2}k_1) - (x_0 + \frac{h}{2})^2 + 1 \\ &= (0.5 + \frac{0.2}{2} \cdot 1.5) - (0 + \frac{0.2}{2})^2 + 1 = 1.64 \end{aligned}$$

$$\begin{aligned} y_1 &= y_0 + h \cdot k_2 \\ &= 0.5 + 0.2 \cdot 1.64 = 0.828 \end{aligned}$$

$$\begin{aligned} i = 1: \quad k_1 &= f(x_1, y_1) = y_1 - x_1^2 + 1 \\ &= 0.828 - 0.2^2 + 1 = 1.788 \end{aligned}$$

$$\begin{aligned} k_2 &= f(x_1 + \frac{h}{2}, y_1 + \frac{h}{2}k_1) = (y_1 + \frac{h}{2}k_1) - (x_1 + \frac{h}{2})^2 + 1 \\ &= (0.828 + \frac{0.2}{2} \cdot 1.788) - (0.2 + \frac{0.2}{2})^2 + 1 = 1.9168 \end{aligned}$$

One-Step Method

Example 07-05: Approximate the numerical solution of the following initial value problem using **the midpoint method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2$$

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The Midpoint Method:

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1)$$

$$y_{i+1} = y_i + h \cdot k_2$$

$$\begin{aligned} i = 0: \quad k_1 &= f(x_0, y_0) = y_0 - x_0^2 + 1 \\ &= 0.5 - 0^2 + 1 = 1.5 \end{aligned}$$

$$\begin{aligned} k_2 &= f(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_1) = (y_0 + \frac{h}{2}k_1) - (x_0 + \frac{h}{2})^2 + 1 \\ &= (0.5 + \frac{0.2}{2} \cdot 1.5) - (0 + \frac{0.2}{2})^2 + 1 = 1.64 \end{aligned}$$

$$\begin{aligned} y_1 &= y_0 + h \cdot k_2 \\ &= 0.5 + 0.2 \cdot 1.64 = 0.828 \end{aligned}$$

$$\begin{aligned} i = 1: \quad k_1 &= f(x_1, y_1) = y_1 - x_1^2 + 1 \\ &= 0.828 - 0.2^2 + 1 = 1.788 \end{aligned}$$

$$\begin{aligned} k_2 &= f(x_1 + \frac{h}{2}, y_1 + \frac{h}{2}k_1) = (y_1 + \frac{h}{2}k_1) - (x_1 + \frac{h}{2})^2 + 1 \\ &= (0.828 + \frac{0.2}{2} \cdot 1.788) - (0.2 + \frac{0.2}{2})^2 + 1 = 1.9168 \end{aligned}$$

$$\begin{aligned} y_2 &= y_1 + h \cdot k_2 \\ &= 0.828 + 0.2 \cdot 1.9168 = 1.21136 \end{aligned}$$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2$$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2$$

Exact Solution: $y(x) = (x + 1)^2 - \frac{1}{2}e^x$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2$$

Exact Solution: $y(x) = (x + 1)^2 - \frac{1}{2}e^x$

$$f(x, y) = y - x^2 + 1, N = \frac{2 - 0}{0.2} = 10, x_0 = 0, y_0 = 0.5$$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2$$

Exact Solution: $y(x) = (x + 1)^2 - \frac{1}{2}e^x$

$$f(x, y) = y - x^2 + 1, N = \frac{2 - 0}{0.2} = 10, x_0 = 0, y_0 = 0.5$$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

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Fourth Order Explicit Runge-Kutta Method:

$$k_1 = f(x_i, y_i)$$

$$k_3 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_2\right)$$

$$k_2 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right)$$

$$k_4 = f(x_i + h, y_i + hk_3)$$

$$y_{i+1} = y_i + \frac{h}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4)$$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

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Fourth Order Explicit Runge-Kutta Method:

$$k_1 = f(x_i, y_i) \qquad k_3 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_2\right)$$

$$k_2 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right) \qquad k_4 = f(x_i + h, y_i + hk_3)$$

$$y_{i+1} = y_i + \frac{h}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4)$$

$i = 0:$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2$$

Exact Solution: $y(x) = (x + 1)^2 - \frac{1}{2}e^x$

$$f(x, y) = y - x^2 + 1, N = \frac{2 - 0}{0.2} = 10, x_0 = 0, y_0 = 0.5$$

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$$y_{i+1} = y_i + \frac{h}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\begin{aligned} i = 0: \quad k_1 &= f(x_0, y_0) = y_0 - x_0^2 + 1 \\ &= 0.5 - 0^2 + 1 = 1.5 \end{aligned}$$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2 \quad \text{Exact Solution: } y(x) = (x + 1)^2 - \frac{1}{2}e^x$$

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Fourth Order Explicit Runge-Kutta Method:

$$\begin{aligned} k_1 &= f(x_i, y_i) & k_3 &= f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_2\right) \\ k_2 &= f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right) & k_4 &= f(x_i + h, y_i + hk_3) \\ y_{i+1} &= y_i + \frac{h}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4) \end{aligned}$$

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One-Step Method

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One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2 \quad \text{Exact Solution: } y(x) = (x + 1)^2 - \frac{1}{2}e^x$$

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$$\begin{aligned} k_4 &= f(x_1 + h, y_1 + hk_3) = (y_0 + hk_3) - (x_0 + h)^2 + 1 \\ &= (0.5 + 0.2 \cdot 1.654) - (0 + 0.2)^2 + 1 = 1.7908 \end{aligned}$$

One-Step Method

Example 07-06: Approximate the numerical solution of the following initial value problem using **fourth order explicit Runge-Kutta method** for $0 \leq x \leq 2$ with step size $h = 0.2$.

$$y' = y - x^2 + 1, y(0) = 0.5; \quad 0 \leq x \leq 2 \quad \text{Exact Solution: } y(x) = (x + 1)^2 - \frac{1}{2}e^x$$

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$$\begin{aligned} y_1 &= y_0 + \frac{h}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4) \\ &= 0.5 + \frac{0.2}{6} \cdot (1.5 + 2 \cdot 1.64 + 2 \cdot 1.654 + 1.7908) = 0.829293 \end{aligned}$$

$$\begin{aligned} k_3 &= f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_2\right) = \left(y_0 + \frac{h}{2}k_2\right) - \left(x_0 + \frac{h}{2}\right)^2 + 1 \\ &= \left(0.5 + \frac{0.2}{2} \cdot 1.64\right) - \left(0 + \frac{0.2}{2}\right)^2 + 1 = 1.654 \end{aligned}$$

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